Introduction to Linear and Abstract Algebra

1. A 1×5 rectangle is split into five unit squares (cells) numbered 1 through 5 from left to right. A frog starts at cell 1. Every second it jumps from its current cell to one of the adjacent cells. The frog makes exactly 14 jumps. How many paths can the frog take to finish at cell 5?

2. For a sequence containing B's and R's, in each move you can insert or remove "BB", "RRR" or "RBRB" anywhere. If you begin with RB, prove that you can't get BR after some moves.

3. Consider any binary word (i.e. string that contain only 0's and 1's). It can be transformed by inserting, deleting or appending any word *XXX*, *X* being any binary word. If a binary word can be changed to another, they are in the same equivalent class; otherwise, they are not. Find the amount of equivalent classes.

4. There are 2021 coins in a row, initially the n-th coin is showing tails, the others are showing heads. In each move you can choose a coin showing tails, and flip all its neighboring coins.

Find all $1\leq n\leq 2021$ such that you can let all coins show tails according to this rule.

5. Let $\varepsilon = \mathrm{e}^{\frac{2\pi \mathrm{i}}{7}}$, the set $S=\{1,\varepsilon,\varepsilon^2,\varepsilon^3,\varepsilon^4,\varepsilon^5,\varepsilon^6\}$. Function $f:S\to S$ takes the form $f(x)=\varepsilon^{\alpha}x^{\beta}$, where $\alpha\in\{0,1,2,3,4,5,6\}$ and $\beta\in\{1,2,3,4,5,6\}$. Let F be the set of such functions. Does there exist a subset G of F satisfying $|F|=2|G|$, and the elements in G are closed under the operation "composition of functions"?

6. Is the alternating A_4 isomorphic to the "affine group" of a ring? What about A_5?

7. For any positive integer n, denote by $S(n)$ its digital sum and $f(n)=(-1)^{S(n)}n^{100}$.

Find the largest power of 5 that divides $\sum_{i=1}^{10^{100}-1} f(i)$.

8. For any positive integer a, denote by $r(a)$ the amount of inversions in its digits (which is the amount of times that a larger digit occurs before a smaller digit, e.g. $r(987654321)=36$). Let S be the set of 9-digit numbers that contain 1 to 9 each once.

Find the smallest positive integer n such that $\sum_{a\in S}(-1)^{r(a)}a^n \neq 0$.

9. Let m and n be positive integers, define:

$f(x)=(x-1)(x^2-1)\cdots(x^m-1)$, $g(x)=(x^{n+1}-1)(x^{n+2}-1)\cdots(x^{n+m}-1)$.

Prove that there exists a polynomial $h(x)$ of degree mn such that $f(x)h(x)=g(x)$ and its $mn+1$ coefficients are positive integers.

10. Let p be a prime number and $n\geq p$ be an integer. Let M be a set of n integers. Denote by f_k the amount of k-subsets of M whose element sum is divisible by p. Prove that $\sum_{k=0}^{n}(-1)^k f_k$ is divisible by p.

11. Given a finite set S, $P(S)$ denotes the set of all the subsets of S. For any function f from $P(S)$ which takes real values, prove that $\sum_{A\in P(S)}\sum_{B\in P(S)} f(A)f(B)2^{|A\cap B|}\geq 0$.

12. In each cell of a 4×11 grid, the number 1 is written. A *move* consists of choosing a positive integer k and a cell, and then multiplying the numbers in that cell and its neighbors by k. Is it possible, after a finite number of moves, for every cell on the grid to contain the number 2025^{2026}?

Note: Two cells are considered neighbors if they share a side.